

Programme	BSc Statistics				
Course Code	ACT1MN101 (P)				
Course Title	Actuarial mathematics I				
Type of Course	Minor				
Semester	I				
Academic Level	100 - 199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Basic knowledge of rates of interest, arithmetic skills, Basic computer skills				
Course Summary	The aim of this course is to provide a grounding in the principles of modelling as applied to actuarial work				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Describe how to use a generalized cash flow model in financial transaction.	U	F	Instructor-created exams / Quiz
CO2	Apply different kinds of interest rates expressed in different time periods	Ap	P	Instructor-created exams / Home Assignments
CO3	Recall and use the more important compound interest functions including annuities certain.	Ap	P	Seminar Presentation / Group Tutorial Work
CO4	Describe how a loan may be repaid by regular instalments of interest and capital.	U	C	Instructor-created exams / Home Assignments
CO5	Recall how to take into account time value of money using the concepts of compound interest and discounting.	R	P	One Minute Reflection Writing assignments
CO6	Solving cash flow models with sample data with spread sheet	Ap	P	Viva-Voce/Practical Assignment
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hours (45 +30)	Marks (70)
I	Cash flow models		8	10
	1	Cash flow process, Examples of cash flow scenarios	2	
	2	A zero-coupon bond, A fixed interest security	2	
	3	An index-linked security, Cash on deposit	2	
	4	An equity, An “interest-only” loan, A repayment loan (or mortgage)	2	
	Sections from References: Unit 1: 3.1,3.2 [Ref 1] Unit 2: 3.2.1,3.2.2 [Ref 1] Unit 3: 3.2.3,3.2.4 [Ref 1] Unit 4: 3.2.5,3.2.7,3.2.8 [Ref 1]			
II	The time value of money		9	15
	5	Interest, Simple interest, Compound (effective) interest	2	
	6	Accumulation factors, The principle of consistency, Present values	2	
	7	Discount rates, Simple discount, Compound (effective) discount, Discount factors, Effective rates of interest and discount	3	
	8	Equivalent rates	2	
	Sections from References: Unit 5: 4.1,4.1.1,4.1.2, [Ref 1] Unit 6: 4.1.3,4.1.4,4.2 [Ref 1] Unit 7: 4.3(4.3.1 to 4.3.3),4.4 - [Ref 1] Unit 8: 4.5- [Ref 1]			
III	Interest Rates		12	25
	9	Nominal rates of interest	2	
	10	Accumulating and discounting using nominal interest rates	2	
	11	Nominal rate of discount	2	
	12	Accumulating and discounting using nominal discount rates	2	
	13	The force of interest(Concept only)	1	

	14	Accumulating and discounting using the force of interest	2	
	15	Relationship between force of interest, effective rate of interest, effective rate of discount, and present value function, v.	1	
	Sections from References: Unit 9: 5.1,5.1.1([Ref 1] Unit 10: 5.1.2 [Ref 1] Unit 11: 5.1.3 [Ref 1] Unit 12: 5.1.4 [Ref 1] Unit 13: 5.2 [Ref 1] Unit 14: 5.2.2 [Ref 1] Unit 15: 5.2.2 [Ref 1]			
IV	Level annuities and Equations of values		16	20
	16	Present values, Payments made in arrear, Payments made in advance	2	
	17	Accumulations of annuities	2	
	18	Continuously payable annuities	3	
	19	Annuities payable pthly- Accumulations	2	
	20	Perpetuities, Perpetuities payable pthly, Deferred annuities	2	
	21	Equations of value, Solving for an unknown quantity	2	
	22	Solving for the timing of a payment (n), Solving for the interest rate (i)	3	
	Sections from References: Unit 16: 8.1(8.1.1 to8.1.2) [Ref 1] Unit 17: 8.2 [Ref 1] Unit 18: 8.3[Ref 1] Unit 19: 8.4 (8.4.1 to 8.4.2) [Ref 1] Unit 20: 8.6,8.7,8.7.1[Ref 1] Unit 21: 10.1&10.1.2[Ref 1] Unit 22: 10.1.2 [Ref 1]			
V	PRACTICUM		30	
	Do practice problems in spreadsheet from any 5 units of the given list and one additional problem decided by the teacher-in-charge, related to the content of the course. Other units listed here may be used as demonstrations of the concepts taught in the course.			
	1	Simple interest and compound interest		
	2	Present value		
	3	Accumulated value		
	4	Force of interest		

	5	Annuities		
	6	Relationship between various interest rates		
	7	Perpetuity		
	8	Equations of values		

Books and References:

6. Institute of Actuaries Act Ed. Study materials CM1
7. McCutcheon, J.J., Scott William (1986): An introduction to Mathematics of Finance
8. Butcher, M.V., Nesbit, Cecil. (1971) Mathematics of compound interest, Ulrich's Books
9. David Promislow, S. (2014), Fundamentals of Actuarial Mathematics, John Wiley & Sons.
10. Newton L Bowers, et al (1997): Actuarial Mathematics, The Society of Actuaries, 2nd Ed
11. Shailaja R. Deshmukh- Actuarial Statistics-an introduction using R, 3rd Ed.

Mapping of COs with PSOs and POs:

	PSO 1	PSO 2	PSO 3	PSO 4	PSO 5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	2	1	3	1	1	-	2	-	3	-	-	-
CO 2	3	3	1	2	1	-	3	1	2	1	-	-
CO 3	1	2	-	2	-	-	2	-	2	-	-	-
CO 4	3	2	-	1	-	-	3	-	2	2	-	-
CO 5	3	2	-	-	-	-	3	-	2	-	-	-
CO 6	1	1	2	-	3	3	2	2	1	-	2	2

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics: